

Presidency

School of Artificial Intelligence and Advanced Computing



CSS7102- Mini Project Review-2

AI Assistant for Policy Document Comparison and Impact Analysis

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Abstract

ClauseLens is an AI-powered document comparison system that analyzes two versions of policy or contract documents to identify clause-level changes and assess their impact. The system combines semantic retrieval, clause alignment, deontic reasoning, and a multi-agent framework consisting of an Impact Agent and a Critic Agent to generate accurate, evidence-backed insights. By highlighting affected stakeholders, change severity, and supporting evidence, ClauseLens enables efficient policy review, contract analysis, and compliance monitoring.

Objectives

- To develop an AI-powered system for comparing two versions of policy or contract documents at the clause level.
- To automatically identify added, deleted, and modified clauses using semantic and structural document analysis techniques.
- To extract and analyze legal obligations, permissions, conditions, and exceptions through deontic reasoning.
- To assess the impact of policy changes on different stakeholders and classify the severity and direction of those changes.
- To implement a multi-agent framework consisting of an Impact Agent and a Critic Agent for improved accuracy and reliability.

Proposed Method and Feasibility Study

- Develop **ClauseLens**, an AI-powered system for comparing two versions of policy or contract documents.
- Parse and preprocess documents using **hierarchical chunking** while preserving document structure.
- Use a **hybrid retrieval mechanism** combining BM25, semantic embeddings, and ChromaDB to locate relevant clauses.
- Identify **added, deleted, and modified clauses** through semantic similarity and Natural Language Inference (NLI).
- Extract legal elements such as **obligations, permissions, prohibitions, conditions, and exceptions** using deontic reasoning.
- Analyze stakeholder impact, severity, and change direction using an **Impact Agent**.

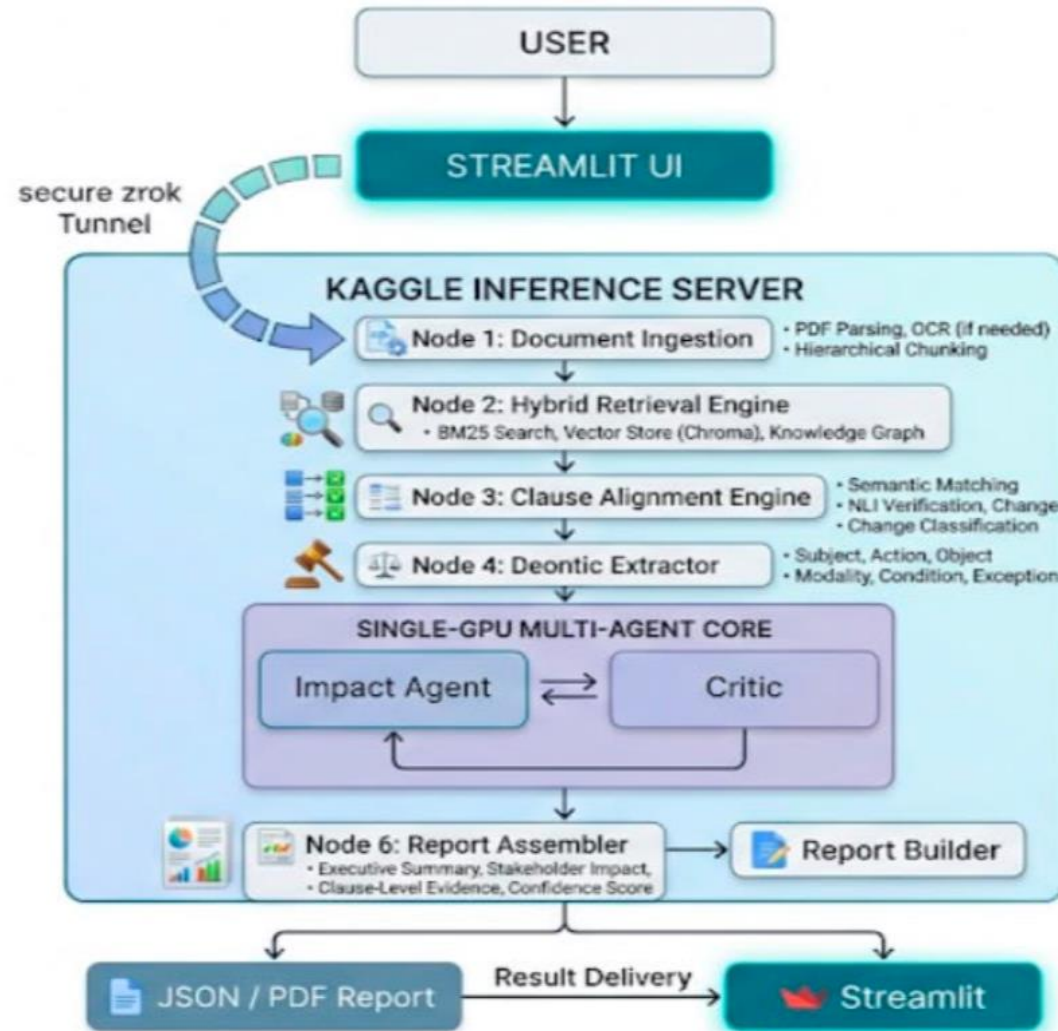


Proposed Method and Feasibility Study (contd...)

- Utilizes open-source technologies such as **Python, PyTorch, Hugging Face, ChromaDB, and Streamlit**.
- Leverages **Gemma models** for legal document understanding and impact analysis.
- Uses the **CUAD dataset** for clause extraction and legal-domain training.
- Kaggle's GPU environment provides sufficient resources for model training and inference.
- Required tools and frameworks are mature, well-documented, and widely adopted.



Architecture Diagram



Hardware and Software Details

Category	Component	Purpose
OS / Platform	Kaggle Notebooks (cloud)	Hosts GPU training & inference sessions
Programming Language	Python 3.x	Core development language
ML Framework	PyTorch	Tensor operations, GPU acceleration
Model Hub	HuggingFace	Downloading Gemma 4 (E2B/E4B) base models & adapters
Fine-tuning	QLoRA	Lightweight adapter-based fine-tuning on CUAD
Quantization	BitsAndBytes	Compressing models to fit within GPU memory
Vector Database	ChromaDB	Storing embeddings for hybrid (BM25 + semantic) retrieval
Document Parsing	PyMuPDF	Extracting text from PDF/HTML policy & contract documents
Frontend	Streamlit	Web UI for uploading documents & viewing reports
Networking	Zrok (secure tunneling)	Connecting local Streamlit UI to Kaggle backend
Dataset	CUAD	510 contracts, 13,000+ clause annotations, 41 clause categories

Hardware and Software Details (Contd...)

Resource	Specification	Why It's Needed
GPU	Dual NVIDIA T4 (15 GB VRAM each, 30 GB total)	Runs Impact Agent (GPU 0) and Critic Agent (GPU 1) concurrently — true parallel multi-agent inference
System RAM	30 GB	Holds full CUAD dataset, Chroma vector store, and large (50+ page) documents in memory simultaneously
Disk Storage	57 GB	Checkpointing pipeline state after every node; enables resume-on-crash without reprocessing
Model Size	Gemma 4 E2B (2 GB) / E4B (4 GB)	Small footprint allows two independent model instances to fit on separate GPUs
Client Device	Laptop (CPU-only)	Runs Streamlit UI locally; only sends/receives files via tunnel — no GPU needed client-side



Timeline of the Project (Gantt Chart)

Task / Milestone	Aug W1	Aug W2	Aug W3	Aug W4	Sep W1	Sep W2	Sep W3	Sep W4	Oct W1	Oct W2	Oct W3	Oct W4
Literature Review & Problem Finalization												
CUAD Dataset Study & Ontology Dictionary Creation												
Mining Real Version Pairs (SEC EDGAR - Strategy A)												
Generating Synthetic Version Pairs (Strategy B)												
Kaggle Training Session Setup (Gemma 4 + QLoRA)												
Fine-tuning & Adapter Training on CUAD Categories												
Node 1-2: Document Ingestion, Chunker & Vector Store												
Node 3-4: Alignment Engine & Deontic Cue Extractor												
Node 5: Parallel Multi-Agent Core (Impact + Critic)												
Node 6: Report Assembler & Streamlit UI Integration												
End-to-End Pipeline Testing & Debugging												
Documentation & Final Review Preparation												

References (IEEE Paper format)

- A. Suresh, M. N. H. N. M. N. et al., "LegDiff: Fine-Grained Semantic Comparison of Legal Documents Using Large Language Models," in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics: Student Research Workshop (ACL SRW)*, 2026.
- M. Umer, M. M. Hassan, et al., "Evaluating Knowledge-Based Approaches for Legal Text Analysis: A Benchmark Study," *Machine Learning with Applications*, vol. 17, 2026.
- K. Chalkidis, I. Androutsopoulos, N. Aletras, et al., "Natural Language Processing for the Legal Domain," *ACM Computing Surveys*, vol. 56, no. 3, pp. 1-38, 2023.
- S. Mandal, N. K. Sahu, et al., "Legal Case Document Similarity: You Need Both Network and Text," *arXiv preprint arXiv:2209.12474*, 2022.
- K. Chalkidis, I. Androutsopoulos, M. Malakasiotis, et al., "LexGLUE: A Benchmark Dataset for Legal Language Understanding in English," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022, pp. 4310-4330.



Thank
You!

